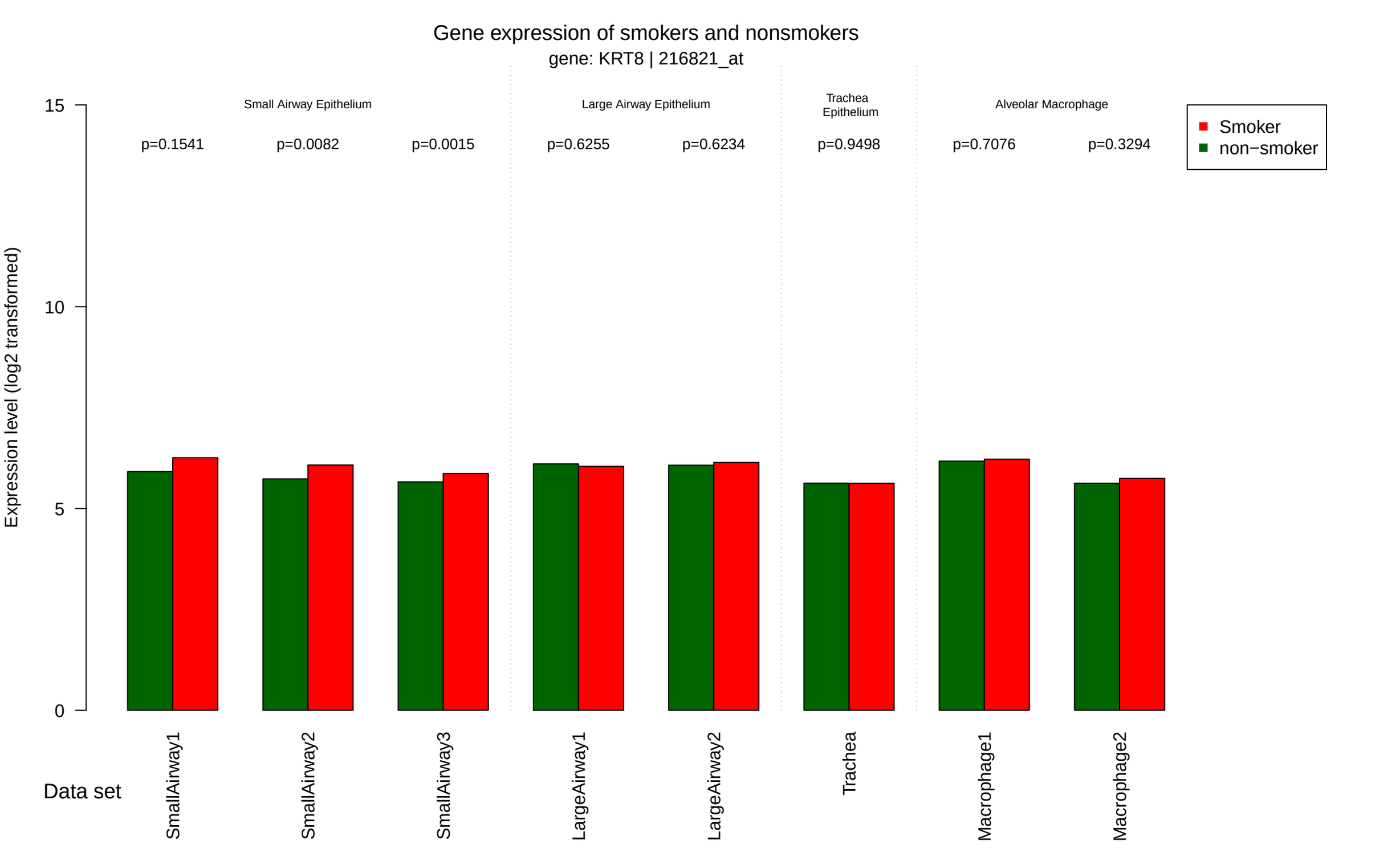
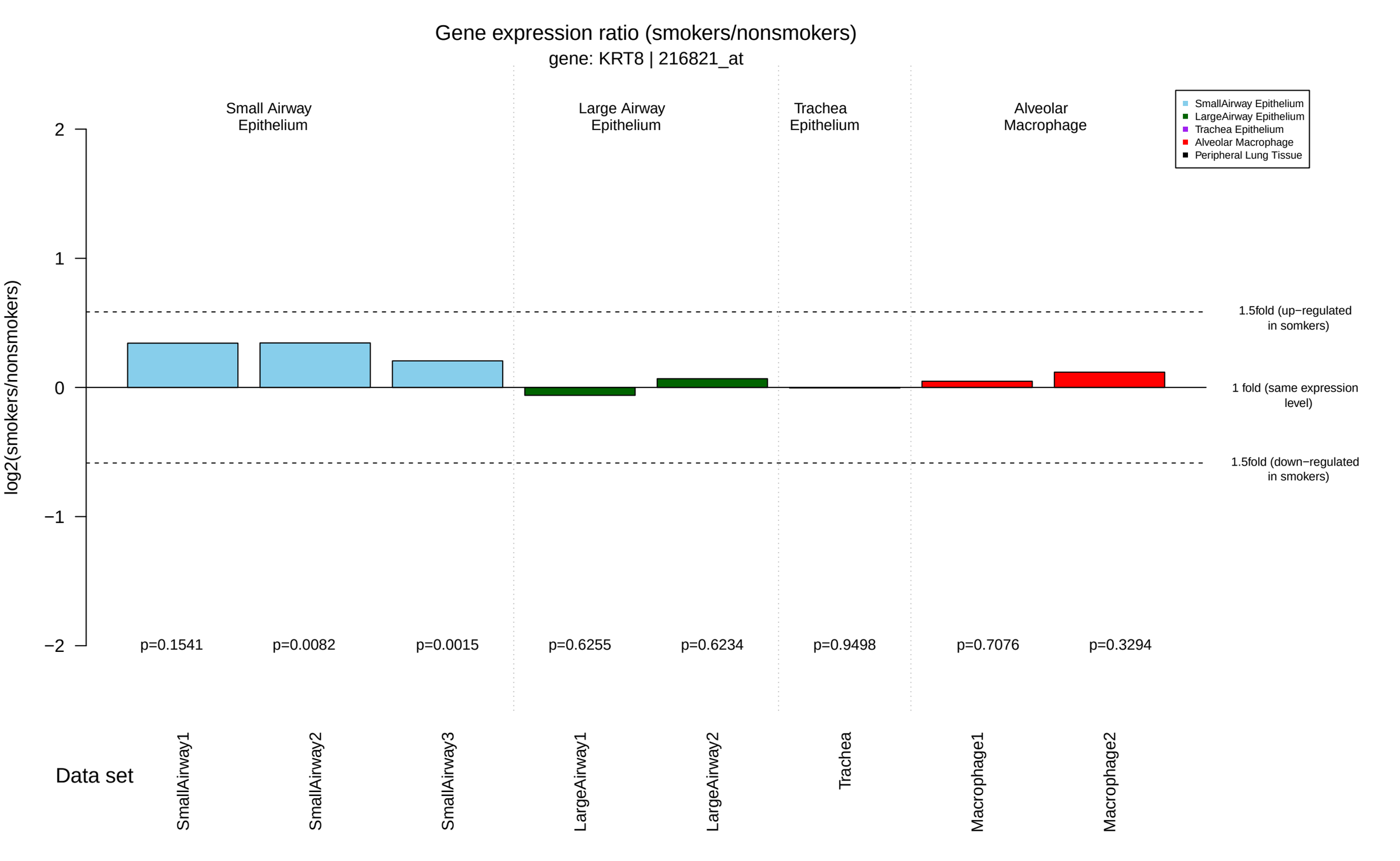
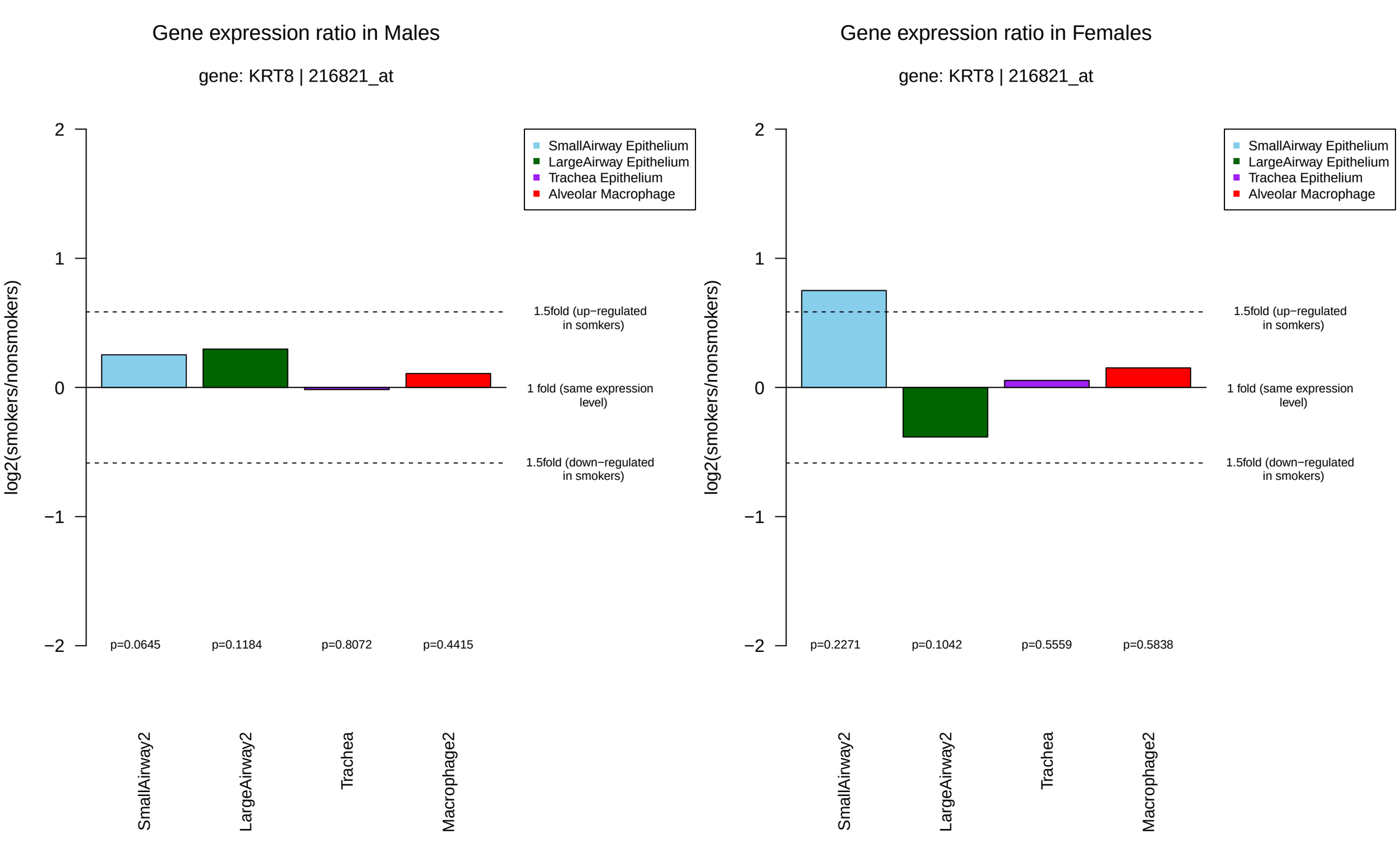
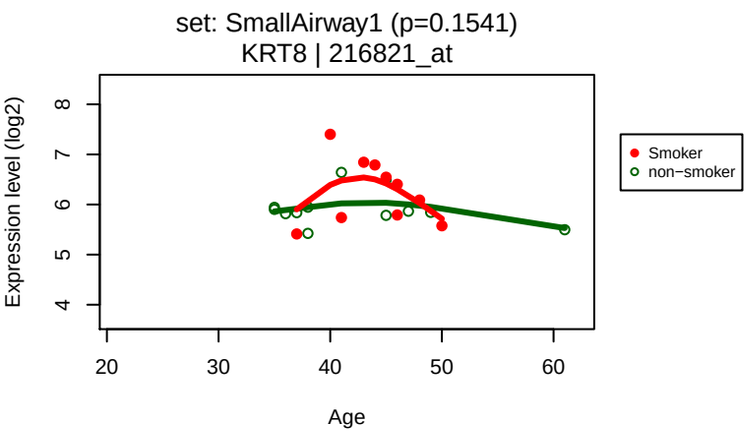
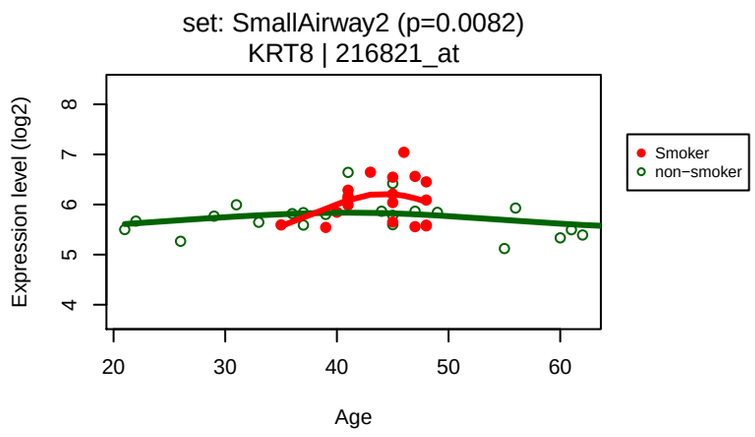
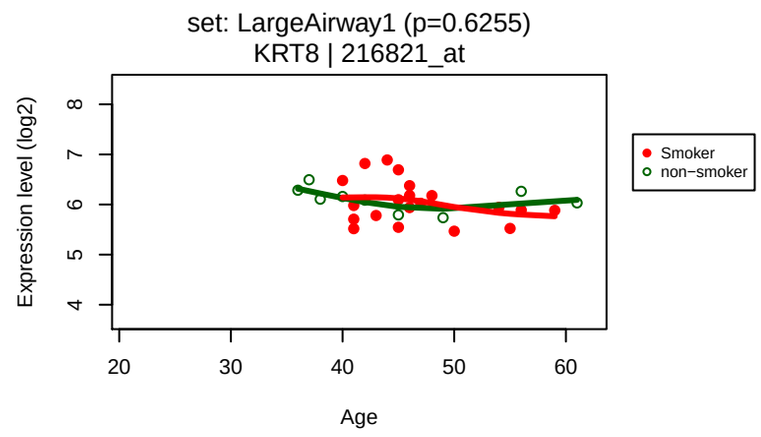

Gene expression of smokers and nonsmokers
gene: KRT8 | 216821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1541
p=0.0082
p=0.0015
p=0.6255
p=0.6234
p=0.9498
p=0.7076
p=0.3294
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

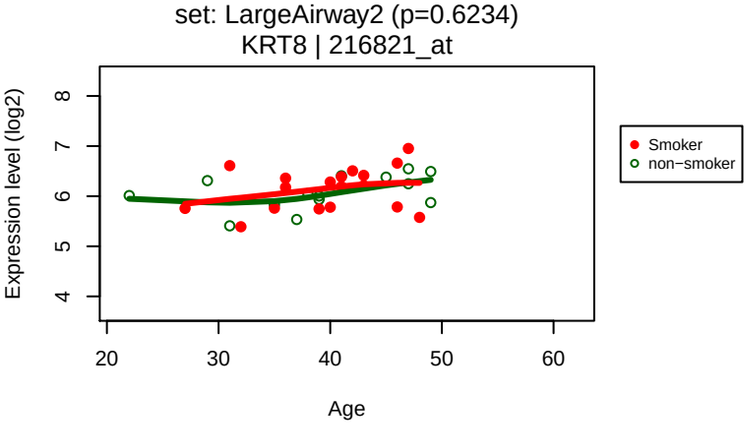

Gene expression ratio (smokers/nonsmokers)
gene: KRT8 | 216821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1541
p=0.0082
p=0.0015
p=0.6255
p=0.6234
p=0.9498
p=0.7076
p=0.3294
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

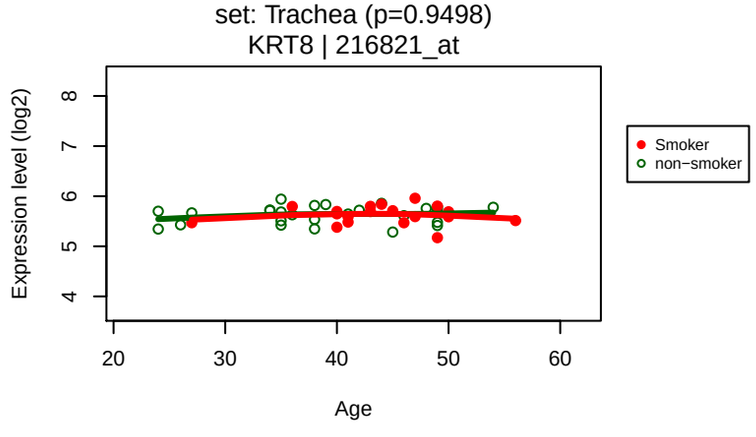

Gene expression ratio in Males
gene: KRT8 | 216821_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0645
p=0.1184
p=0.8072
p=0.4415
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRT8 | 216821_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2271
p=0.1042
p=0.5559
p=0.5838
SmallAirway2
LargeAirway2
Trachea
Macrophage2

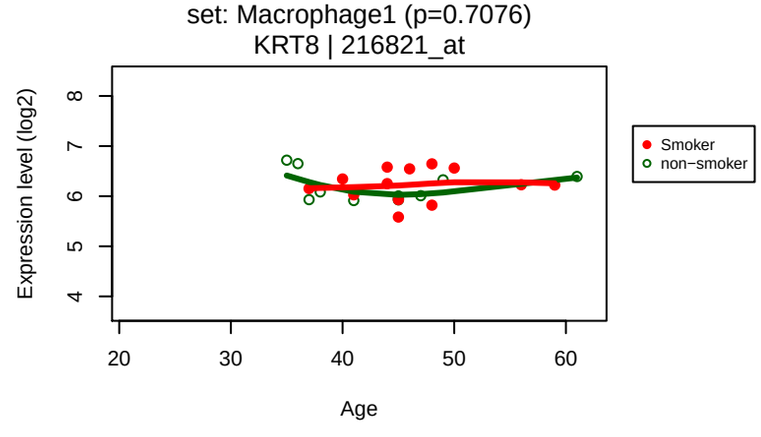

set: SmallAirway1 (p=0.1541)
KRT8 | 216821_at
Expression level (log2)
Age
Smoker
non−smoker

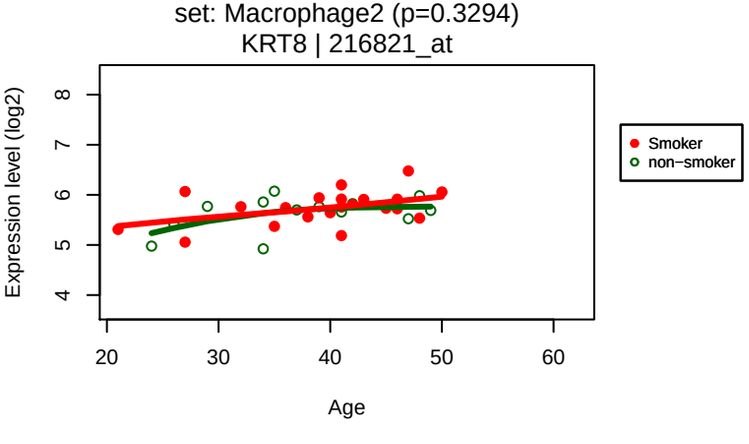

set: SmallAirway2 (p=0.0082)
KRT8 | 216821_at
Expression level (log2)
Age
Smoker
non−smoker

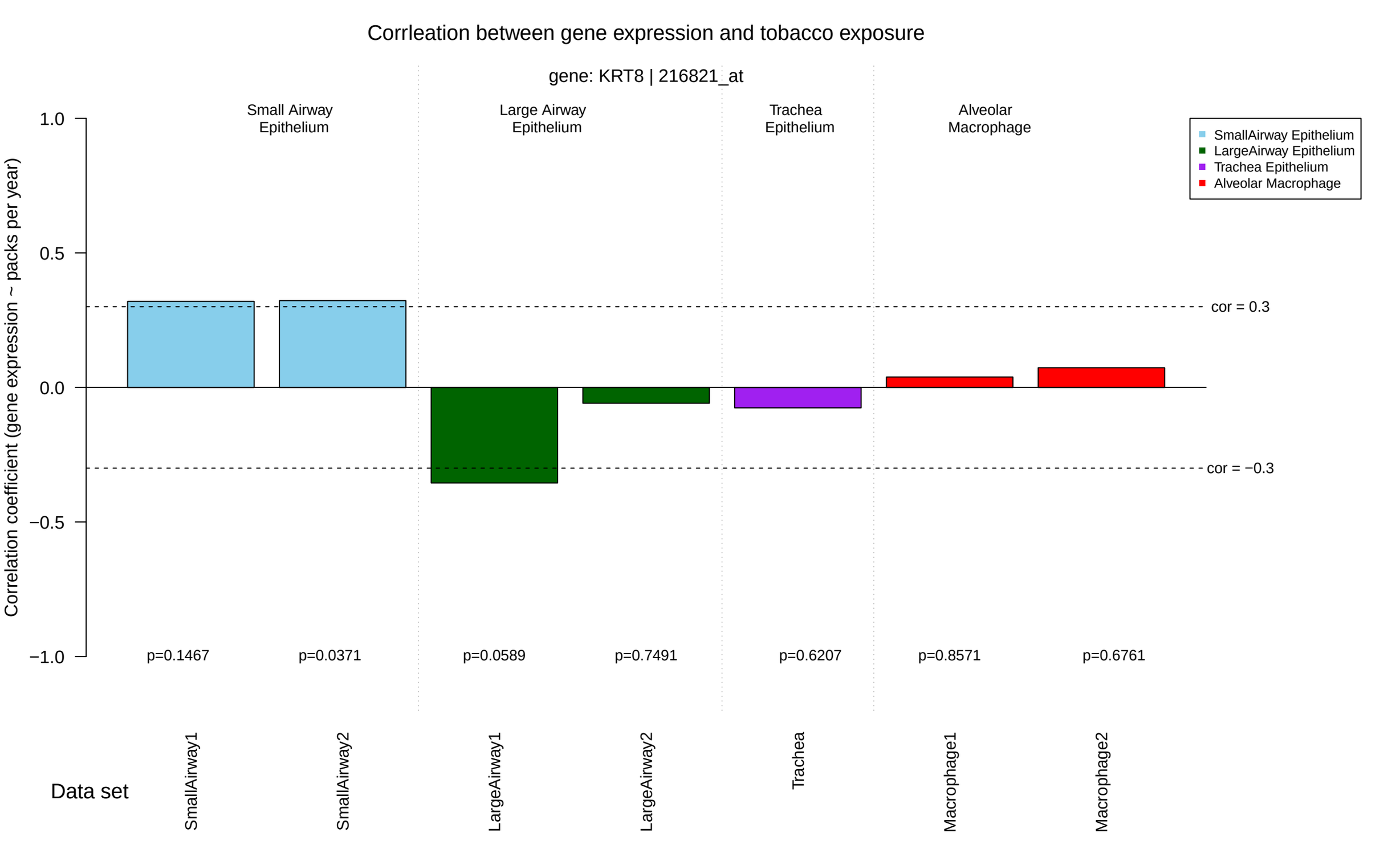

set: LargeAirway1 (p=0.6255)
KRT8 | 216821_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6234)
KRT8 | 216821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9498)
KRT8 | 216821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7076)
KRT8 | 216821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3294)
KRT8 | 216821_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRT8 | 216821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1467
p=0.0371
p=0.0589
p=0.7491
p=0.6207
p=0.8571
p=0.6761
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2